

Hacettepe University | Computer Engineering CMP 720 EMBEDDED SYSTEMS DESIGN

Spring 2026

Midterm Project Peer Evaluation Form

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Instructions

For each project, briefly summarize the work and provide your evaluation **individually (not as a group)**. Your response should reflect your understanding of the problem, methodology, and evaluation plan, and include critical insights, strengths, limitations, and possible improvements. Skip your own group's evaluation and simply state that it is your group.

Students are required to evaluate only the projects presented within their own class group (Hacettepe or ASELSAN).

Guiding points (address in one coherent paragraph):

- What problem is being solved, why is it important, and what are the most important embedded system design considerations for this problem (e.g., timing, reliability, energy, safety, security, scalability, etc.)?
- What is the main idea of the proposed methodology?
- What do you think are the strengths and limitations of the approach?
- Is there anything you would improve or do differently?
- What is the most critical or challenging component?
- Is the evaluation plan sufficient? What, if anything, is missing?

Group 1

Umut Kalman, Eren Karadağ

Project Title: *Context-Aware Deadline Scheduling Under Memory Contention for Embedded Real-Time Systems*

This project focuses on deadline misses caused by memory contention in embedded real-time systems. I think the problem is important because a task can look schedulable on paper, but still miss its deadline when several memory-heavy tasks run together. Their idea is to improve Earliest Deadline First scheduling with a simple memory penalty model and a slack check. I like that the solution is lightweight and does not require changing the kernel deeply. The main weak point is that the memory model is still simplified, so it may not fully represent cache and bus behavior on real hardware. The most difficult part seems to be measuring memory intensity correctly. Their metrics are meaningful, but I would like to see more tasks and a larger test case to understand scalability better.

Group 2

Yusuf Çağrı Ögüt

Project Title: *Hybrid Cellular-LoRa Network Based Agriculture Monitoring and Control Application***Group 3**

Yusuf Tahir Orhan

Project Title: *Implementation of DeFT: A Deadlock-Free and Fault-Tolerant Routing Algorithm*

This project studies routing problems in 2.5D chiplet networks, especially deadlocks and faulty vertical links. The topic is relevant because communication reliability becomes very important when systems are divided into chiplets. The DeFT method uses virtual network rules for deadlock freedom and lookup tables for fault-aware routing. I think the strongest part is that the design tries to keep the hardware cost low while still keeping packets reachable under link faults. The main limitation is that the work is mostly an implementation and replication of an existing algorithm, so the novelty is limited. The lookup table mechanism is probably the most critical part, especially when the network becomes larger. The evaluation plan is good, but comparing with more routing algorithms would make the results stronger.

Group 4

Mehmet Yusuf Sezgi, Rana Elif Albayrak

Project Title: *Real-Time CAN Decoder*

This project aims to decode vehicle CAN traffic in real time on a small microcontroller. I find the project useful because CAN buses can produce dense traffic, and losing frames may make the decoded data unreliable. Their solution is a pipeline with acquisition, buffering, processing, and output stages on an STM32F411 using Zephyr RTOS. The two-stage buffering idea is a good design choice because it separates fast CAN reception from slower SD card writes. The biggest limitation is that they do not have a DBC file, so the signal decoding depends on experimental observations and may not generalize to other vehicles. The most critical part is handling SPI sharing between the CAN module and the SD card. The planned tests are reasonable, but I would expect real timing results and frame loss measurements in the final version.

Group 5

Metin Irkıcıatal, Anıl Arda

Project Title: *Edge-AI Based Smart Access Control System*
Group 6

Oğulcan Uğuroğlu, Barış Büyükyılmaz

Project Title: *Contention-Aware Duplication Scheduling for NoC Systems*

Group 7

Ahmet Yiğitalp Demirci

Project Title: *Real-Time Energy-Constrained UAV Optimization*

Group 8

Sami Enes Yılmaz, Kadir Emre Avcı

Project Title: *SmartAquaria System*

This project uses an aquarium control system to show a portable and testable embedded software architecture. The main problem is that embedded code often becomes too dependent on specific hardware, which makes testing and reuse difficult. Their layered C++ design with abstract interfaces is a strong approach because the same logic can work with real hardware or a software simulation. I also like the single-threaded event-based design, since it avoids many concurrency problems. The limitation is that some parts, such as the feeding actuator, are still mocked and the system is not fully validated on real hardware. The most challenging point is keeping the layer boundaries clean as the project grows. The simulation results are useful, but physical tests and power measurements would improve the evaluation.

Group 9

Mehmethan Ayrim

Project Title: *Energy-Aware Mixed-Criticality Scheduling*

Group 10

İrem Gül Subaşı

Project Title: *Fault Injection and Resilience Profiling of Lightweight Transformers*

Group 11

Süleyman Ege Karakaya

Project Title: *Adaptive Data-Movement Strategy for Crypto Accelerators*

Group 12

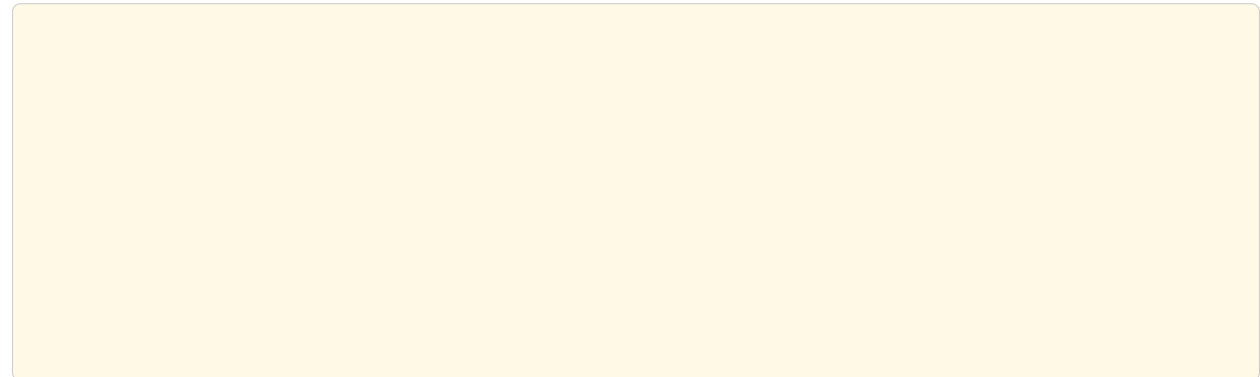
Gökçer Erol

Project Title: *Acoustic Localization System for Search and Rescue*

This project tries to locate survivors after building collapses by using acoustic events. The problem is very meaningful because time is critical in search and rescue operations. The system uses ESP32-S3 nodes to capture audio, filter sounds with TinyML, and send timestamped events to a Jetson Nano for Time Difference of Arrival based localization. I think the strongest part is the combination of edge filtering, GNSS time synchronization, and low-bandwidth LoRa communication. The main limitation is the star topology, which may have packet collisions when many nodes are used in the same area. The GNSS pulse-per-second timing looks like the most critical component because localization accuracy depends on it. The evaluation plan is quite clear, but testing in a realistic noisy environment would be especially important.

Group 13

Emre Doğan, Yavuz Selim Kızıltaş

Project Title: *Optimization of Edge AI Pipeline on TDA4VM***Group 14**

Mehmet Yusuf Bilge, Sezil Ağırbaş

Project Title: *Closed-Loop Vestibular Stimulation System*

This project proposes a closed-loop vestibular stimulation system to reduce balance problems and fall risk. I think the safety aspect makes this project especially important because the system directly interacts with a human body. The design uses an STM32F411 with FreeRTOS, IMU sensor fusion, a PID controller, and a current driver for stimulation. The strongest part is the focus on safety, including current limits, monitoring, and watchdog protection. The main limitation is that the evaluation is planned on a mechanical platform, not with real users, so the medical effectiveness cannot be fully proven yet. The Howland current pump seems to be the most critical hardware block because it must provide stable current under changing electrode impedance. The evaluation metrics are detailed, but future human-centered validation would be needed.

Group 15

Ayberk Aygün, Berkay Taşkın

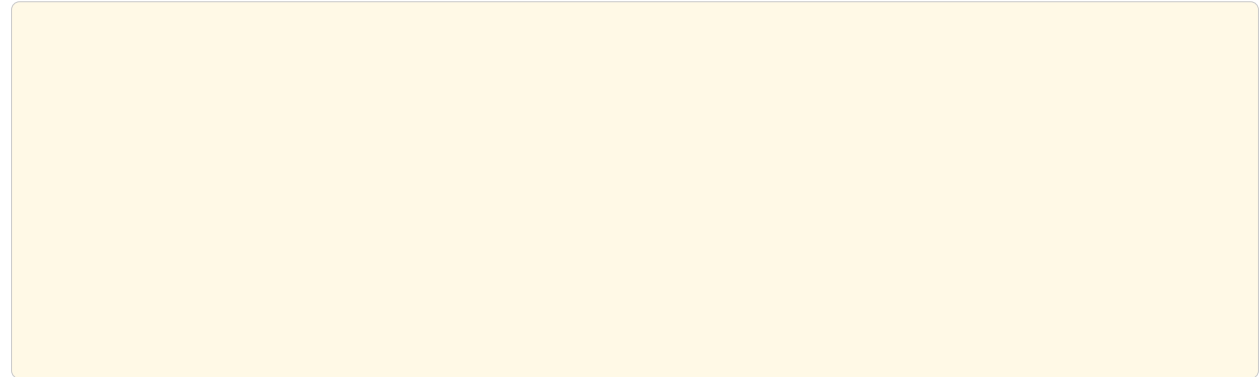
Project Title: *Quadrotor Simulation on NVIDIA Jetson*

This project looks at adaptive PID tuning for quadrotor control on an NVIDIA Jetson platform. The idea is interesting because fixed PID gains may not perform well under changing noise or flight conditions. Their method keeps the control loop on the ARM CPU while the GPU evaluates many candidate PID gains in parallel using Monte Carlo simulations. I think the main strength is using embedded GPU parallelism for online tuning instead of only offline parameter search. The limitation is that the work is simulation-based, so it is not clear how well it would behave on a real drone with sensor delays and actuator limits. The shared memory usage between CPU and GPU is probably the most critical design point. The evaluation constraints are concrete, but real flight or hardware-in-the-loop tests would make the project much more convincing.

Group 16

Yasin Arslan, Süleyman Sertaç Üst

Project Title: *Real-Time Audio Pipeline Analysis on Heterogeneous SoCs*



Group 17

Oğuz Kağan Yağlıoğlu, Emre Eşef Kılıçdoğan

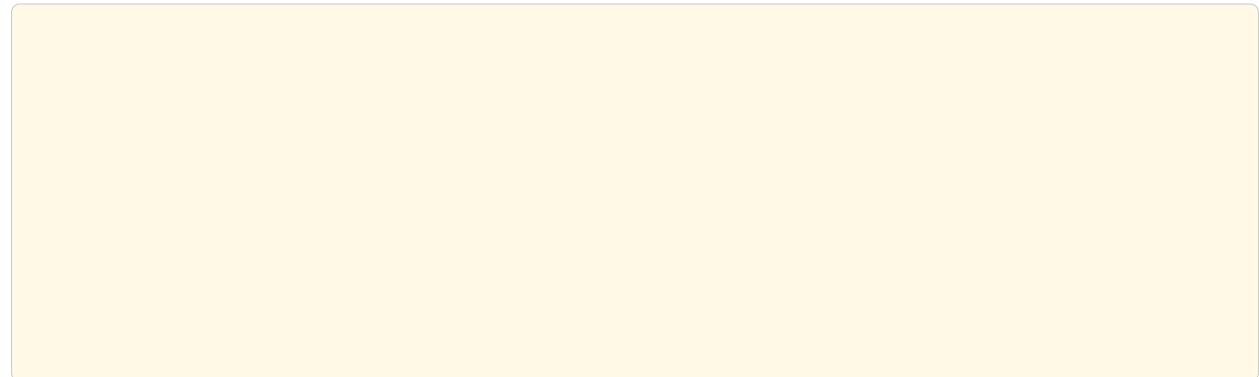
Project Title: *Predictive Maintenance of BLDC Motors via Multi-Sensor Fusion*

This project targets early fault detection in BLDC motors by combining vibration, acoustic, and temperature data. The problem is practical because motor faults can cause production stops and maintenance costs in industrial systems. Their embedded node collects features and sends compact data over LoRaWAN to a fog-layer machine learning classifier. I like that they also include load and speed information, because motor behavior changes with operating conditions. The main limitation is that the reported results are based on a public dataset, not their own hardware setup yet. The STM32WLE5 and sensor integration are the most critical parts of the embedded side. The evaluation looks promising, but testing with the custom PCB and real motor faults is still needed.

Group 18

Hüseyin Eren Doğan

Project Title: *EdgeRAG: Offline RAG on Embedded Systems*



Group 19

Ali Bölücü

Project Title: *Spectral-Seeded NSGA-II for Many-Core Systems*

This project focuses on mapping application tasks onto NoC cores while considering communication cost and temperature. The problem is important because a bad mapping can increase latency and create thermal hotspots. The proposed method uses spectral clustering to create better initial solutions for NSGA-II, so the search algorithm can start from more meaningful placements. I think this is a good idea because optimization algorithms often spend too much time exploring weak initial candidates. The main limitation is the use of an XY routing assumption, which may hide real contention effects inside the network. The Fiedler vector based clustering is the most critical part because poor clustering would reduce the benefit of the method. The evaluation plan uses suitable benchmarks, but I would add Pareto quality metrics such as hypervolume and compare convergence speed clearly.

Group 20

İlter Doğaç Dönmez

Project Title: *Autonomous Obstacle Avoidance Vehicle using LiDAR*

This project is about obstacle avoidance for a small autonomous vehicle using LiDAR. The problem is useful for indoor navigation, where the robot must react quickly and safely to obstacles. The system uses a Raspberry Pi 3, an RPLidar A2, and a motor driver, with a reactive algorithm for movement decisions. I think the LiDAR-only approach is a practical strength because it can work better than cameras in low-light conditions. Using Renode for hardware emulation before real testing is also a nice addition. The main limitation is that the system is mainly designed for indoor and mostly static environments. The RPLidar processing pipeline is the most critical part because sensor latency directly affects driving decisions. The evaluation needs clearer metrics, such as obstacle distance, reaction time, success rate, and test map complexity.